



SDU University

School of Information Technologies and Applied Mathematics

DIPLOMA PROJECT

Machine Learning–Based Prediction System
for Photoelectrochemical Water Splitting Performance

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6B06101 — “Information Systems”

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DIPLOMA PROJECT

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Project Title:

Machine Learning-Based Prediction System for
Photoelectrochemical Water Splitting Performance

Project Tasks:

1. Curate a structured dataset from published PEC literature and implement a physics-based augmentation pipeline.
2. Design and implement a physics-informed feature engineering module.
3. Train and evaluate machine learning models for prediction.
4. Develop a web-based interface for predictions.
5. Analyze results and validate model performance.

Abstract

Photoelectrochemical (PEC) water splitting is one of the most promising pathways for solar hydrogen production, but the experimental search for high-performance photoelectrode materials is slow and resource-intensive. This work presents a full-stack machine learning system that predicts three key performance metrics of PEC cells — Photocurrent Density (mA/cm^2), Solar-to-Hydrogen (STH) efficiency (%), and Hydrogen Evolution Rate ($\mu\text{mol h}^{-1} \text{cm}^{-2}$) — directly from material and experimental parameters.

A curated dataset of 676 entries was assembled from 529 published experimental results, augmented with physics-derived estimates and literature-based synthetic rows to address severe class imbalance in STH and H_2 targets. Twenty physics-informed features were engineered from raw inputs, including bandgap–pH interaction, overpotential proxy, ionic strength proxy, and Arrhenius-like temperature–pH kinetics terms.

An ensemble of XGBoost and Random Forest regressors is trained per target, with hyperparameter selection via randomised search and generalisation measured by 5-fold cross-validation. Best cross-validated R^2 scores are 0.454 for Photocurrent, 0.533 for STH, and 0.366 for H_2 — compared to 0.065, 0.278, and -0.090 before dataset augmentation.

A REST API (Flask) and single-page web interface expose the models for interactive prediction, returning per-target values, model agreement scores, and a composite performance classification (LOW / MEDIUM / HIGH) calibrated to the training distribution.

Keywords: photoelectrochemical water splitting, machine learning, XGBoost, Random Forest, hydrogen production, solar energy, materials informatics, feature engineering, ensemble learning.

Аңдатпа

Фотоэлектрoхимиялық (PEC) судың ыдырауы күн сәулесінен сутегі өндірудің ең перспективалы жолдарының бірі болып табылады. Алайда жоғары өнімді фотоэлектрод материалдарын эксперименттік іздеу баяу және ресурсты қажет ететін процесс болып қалуда.

Осы жұмыста материал мен эксперименттік параметрлер бойынша PEC ұяшықтарының үш негізгі өнімділік көрсеткішін — фототок тығыздығын (mA/cm^2), Күнге-Сутегіге (STH) тиімділігін (%) және сутегінің бөліну жылдамдығын ($\mu\text{mol h}^{-1} \text{cm}^{-2}$) — болжайтын машиналық оқыту жүйесі ұсынылған.

529 жарияланған эксперименттік нәтижеден жасалған 676 жазбадан тұратын деректер жиыны физикалық бағалаулар мен синтетикалық жолдармен толықтырылды. Жиырма физикаға негізделген белгі инженерлік тәсілмен жасалды. XGBoost және Random Forest регрессорларының ансамблі мақсатты айнымалы бойынша дербес оқытылды.

Ең жақсы кросс-валидация R^2 көрсеткіштері: Фототок үшін — 0.454, STH үшін — 0.533, H_2 үшін — 0.366.

Жүйе Flask REST API және интерактивті веб-интерфейс арқылы ұсынылған, болжамдарды, модель келісімділік бағаларын және оқыту таралымына калибрленген өнімділік класификациясын (ТӨМЕН / ОРТАША / ЖОҒАРЫ) қайтарады.

Түйін сөздер: фотоэлектрoхимиялық судың ыдырауы, машиналық оқыту, XGBoost, Random Forest, сутегі өндіру, күн энергиясы, материалдар информатикасы, белгі инженериясы, ансамблдік оқыту.

Аннотация

Фотоэлектрохимическое (ПЕС) расщепление воды является одним из наиболее перспективных методов производства солнечного водорода, однако экспериментальный поиск высокоэффективных материалов для фотоэлектродов остаётся медленным и ресурсоёмким процессом.

В данной работе представлена полноценная система машинного обучения, которая предсказывает три ключевых показателя эффективности ПЕС-ячеек — плотность фототока (mA/cm^2), КПД преобразования солнечной энергии в водород (STH, %) и скорость выделения водорода ($\mu\text{mol ч}^{-1} \text{cm}^{-2}$) — непосредственно по параметрам материала и условиям эксперимента.

Набор данных из 676 записей был сформирован на основе 529 опубликованных экспериментальных результатов и дополнен физически обоснованными оценками и синтетическими строками из литературных источников. Было разработано 20 физически информированных признаков, включая взаимодействие ширины запрещённой зоны с pH, прокси-оценку перенапряжения, прокси ионной силы и кинетические члены.

Ансамбль из регрессоров XGBoost и Random Forest обучается отдельно для каждой целевой переменной с оценкой обобщающей способности методом 5-кратной кросс-валидации. Наилучшие значения R^2 : для фототока — 0.454, для STH — 0.533, для скорости выделения H_2 — 0.366. По сравнению с небогащённым датасетом показатели выросли на 39, 26 и 46 процентных пунктов соответственно.

Система развёрнута в виде REST API (Flask) и одностраничного веб-приложения, возвращающего предсказания, оценки согласованности моделей и классификацию производительности (НИЗКАЯ / СРЕДНЯЯ / ВЫСОКАЯ), калиброванную относительно обучающего распределения.

Ключевые слова: фотоэлектрохимическое расщепление воды, машинное обучение, XGBoost, Random Forest, производство водорода, солнечная энергия, информатика материалов, конструирование признаков, ансамблевое обучение.

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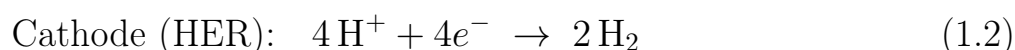
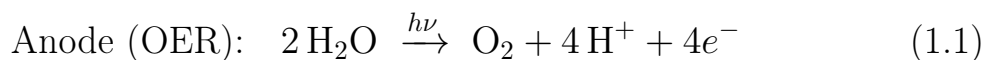
Chapter 1

Introduction

1.1 Problem Background

The global transition away from fossil fuels requires scalable, carbon-free energy carriers. Molecular hydrogen (H_2) is widely regarded as one of the most attractive options: it has the highest gravimetric energy density of any known fuel (120 MJ/kg), produces only water upon combustion, and can be stored and transported using existing infrastructure [1]. Currently, however, more than 95% of global hydrogen is produced by steam methane reforming — a process that emits approximately 10 kg of CO_2 per kilogram of H_2 [2].

Photoelectrochemical (PEC) water splitting offers a pathway to produce green hydrogen directly from sunlight and water using a semiconductor photoelectrode immersed in an aqueous electrolyte [3]. When photons with energy exceeding the semiconductor bandgap are absorbed, they generate electron–hole pairs. The holes oxidise water at the photoanode (oxygen evolution reaction, OER), while the electrons reduce protons at the cathode (hydrogen evolution reaction, HER):



The overall efficiency of this process is quantified by the Solar-to-Hydrogen

(STH) efficiency:

$$\text{STH} = \frac{J_{ph} \times 1.23 \text{ V} \times \eta_F}{P_{in}} \times 100\% \quad (1.3)$$

where J_{ph} is the photocurrent density (mA/cm^2), 1.23 V is the thermodynamic minimum potential for water splitting, η_F is the Faradaic efficiency, and P_{in} is the incident light power density (mW/cm^2).

After decades of research, only a handful of materials come close to the theoretical STH limit for their bandgap [4]. Finding a high-efficiency photoelectrode is not a matter of testing one or two variables — a researcher working with BiVO_4 , for example, must decide on the nanostructure morphology, whether to add a co-catalyst, which electrolyte to use, what pH and applied bias to set, and each choice interacts with the others in ways that are hard to predict in advance. Testing all combinations experimentally is simply not feasible within a typical research timeline.

1.2 Motivation and Relevance

Machine learning has already changed how researchers approach materials discovery in adjacent fields. Battery electrolyte screening, catalysis, and photovoltaics all have ML-assisted workflows now. PEC water splitting does not — and the gap is not because the problem is unimportant, but because the data is hard to collect.

When we went through 529 published papers, we found that fewer than 1 in 10 reported an STH value. Most papers measure photocurrent and stop there, because calculating STH requires additional instrumentation and a more careful experimental setup. The end result is a dataset where the single most important performance metric practically speaking is also the least frequently occurring one in the training data. Standard k-fold cross-validation fails quickly when there are only 48 rows.

This project has a genuinely multidisciplinary nature, which ultimately shaped how I approached the methodology. Electrochemistry allows you to determine which variables to include as features. Materials science informs what we can trust in the experimental data. Machine learning provides the tools to generalise across hundreds of diverse experiments from different laboratories.

Without all three working together, the result is either a physically invalid model, or a scientifically valid one that cannot generalise beyond its training data.

1.3 Aims and Objectives

The main goal of this project is to create a practical application of machine learning. This involves building a machine-learning based system which predicts three different measures — photocurrent density, Solar-to-Hydrogen (STH) efficiency, and Hydrogen Evolution Rate — based upon the provided material and experimental parameters. Once the model was created, we made the developed system available via a user interface, allowing researchers to utilise the machine learning based system.

To achieve our stated goals, we needed to solve multiple issues sequentially:

1. Create a dataset from 529 peer-reviewed papers representing Photoelectrochemical (PEC) water splitting, covering all possible combinations of various materials, structural parameters, and operating conditions, with photocurrent density, Solar-to-Hydrogen (STH) efficiency, and Hydrogen Evolution Rate as predictive variables. Unfortunately, creating the above mentioned dataset proved to be far more challenging than initially anticipated, because most research studies reported only photocurrent density.
2. Develop numerical representations of input values into features by applying established electrochemical relationships. These include the bandgap-pH product as described by the Nernst equation, an overpotential proxy as derived from Butler–Volmer kinetic equations, and hydrogen production rate as derived from Faraday’s law. In doing so, we avoided requiring the machine learning algorithm to derive these same relationships independently.
3. Train separate regression models utilising XGBoost and Random Forest for each predictive variable, evaluate both using 5-fold cross-validation, and combine the results into a single weighted ensemble where weights represent confidence in each model’s output.

4. Make trained models accessible via a Flask REST API and expose them via a React web-based interface, returning predicted values along with a LOW / MEDIUM / HIGH classification of potential success based upon the input configuration entered by users.
5. Compare cross-validated performance metrics to demonstrate that adding physics-derived estimates and synthetic rows to the existing dataset produces significantly improved performance compared to training solely on the original sparse data.

1.4 Structure of the Report

The structure of the report is as follows:

- In Chapter 2, we present all the necessary context to understand our design choices: the physical behaviour of PEC cells; which performance indicators are important and why; and why a tree-based ensemble model has been preferred over a neural network or linear regression.
- In Chapter 3, the content is focused around the methodical development of this project: how the dataset was built from 529 scientific articles; how missing values for STH and H_2 were computed using physics equations; how 20 features were extracted from 8 input parameters; and how the training and prediction pipeline is organised.
- Chapter 4 shows, with examples of three typical material configurations, what the models were able to achieve: cross-validation results before and after augmentation, feature importance analysis, and a system demonstration on representative inputs.
- Chapter 5 summarises the main outcomes, acknowledges where the approach falls short, and outlines the most promising directions for future work.

Chapter 2

Literature Review

2.1 Photoelectrochemical Water Splitting: Fundamentals

2.1.1 Semiconductor Photoelectrodes

The efficiency of that process is determined by several competing elements — the amount of light absorbed by the material, the time it takes for minority charge carriers to reach the surface before they recombine, and the ability of the surface to catalyse the water oxidation reaction [5].

For unassisted water splitting (i.e., no external voltage applied) there are a number of simultaneous requirements that the material must meet [6]: the band gap should fall in approximately the range 1.8–2.2 eV — broad enough to generate sufficient thermodynamic driving forces for both half-reactions, but narrow enough to allow for absorption of a useable fraction of the solar spectrum. The band-edge positions must be positioned such as to bracket the water oxidation and reduction potentials. Finally, the material must remain chemically stable in the electrolyte under prolonged irradiation. Thus far, there exists no single known material that meets all of the above criteria simultaneously; hence, why the experimentally accessible phase-space continues to be so large [4].

2.1.2 Key Performance Metrics

Three metrics are universally reported in PEC literature and constitute the prediction targets in this work.

Photocurrent Density J_{ph} (mA/cm^2) is measured by linear sweep voltammetry under AM 1.5G illumination at a standardised intensity of $100 \text{ mW}/\text{cm}^2$. It reflects the rate of charge transfer to the electrolyte and serves as the primary proxy for combined solar absorption and charge collection efficiency. State-of-the-art values exceed $6 \text{ mA}/\text{cm}^2$ for BiVO_4 [7] and reach above $20 \text{ mA}/\text{cm}^2$ for GaN/InGaN nanowire photoanodes [8].

Solar-to-Hydrogen Efficiency (STH) is the gold standard for overall device efficiency, defined by Equation (1.3). The Shockley–Queisser-analogous theoretical maximum for a single-junction device with $E_g \approx 2.0 \text{ eV}$ is approximately 16.8% [4]. The highest experimentally verified STH values for single-junction photoelectrodes under unassisted, unbiased conditions are approximately 12% for III-nitride tandem structures [8].

Hydrogen Evolution Rate (HER) ($\mu\text{mol h}^{-1} \text{ cm}^{-2}$) is determined by gas chromatography. It is related to photocurrent through Faraday’s law:

$$\text{HER} = \frac{J_{ph} \times 3600}{2F} \times \eta_F \times 10^6 \quad (2.1)$$

where $F = 96\,485 \text{ C/mol}$ is Faraday’s constant. HER is reported less frequently than J_{ph} because it requires a gas-tight electrochemical cell and calibrated gas chromatograph, which explains the severe sparsity of this target in publicly available data.

2.1.3 Principal Material Systems

Table 2.1 summarises the photoelectrode materials covered in this study, their typical bandgap ranges, and representative photocurrent densities.

Table 2.1. Common PEC photoelectrode materials, bandgap ranges, and typical photocurrent performance.

Material	Bandgap (eV)	Typical J_{ph} (mA/cm ²)	Role
BiVO ₄	2.4–2.5	1–7	Photoanode
α -Fe ₂ O ₃	2.0–2.2	0.3–4	Photoanode
WO ₃	2.6–2.8	1–6	Photoanode
TiO ₂	3.0–3.2	0.2–2	Photoanode
ZnO	3.2–3.4	0.5–3	Photoanode
Cu ₂ O	1.9–2.1	1–5	Photocathode
GaN / InGaN	0.7–3.4	5–22	Photoanode
Ta ₃ N ₅	2.0–2.1	0.3–4	Photoanode
Si	1.1	8–30	Photocathode

2.2 Machine Learning for Materials Discovery

2.2.1 General Approaches

Materials informatics applies statistical and machine learning methods to accelerate the discovery of new functional materials [9]. Pilania et al. [10] demonstrated that random forest models trained on DFT-computed features could predict perovskite formation energies with mean absolute errors below 0.1 eV/atom — a task that required only minutes of computation compared to hours of DFT. Isayev et al. [11] introduced universal fragment descriptors capable of predicting band structures across thousands of inorganic compounds.

In the photocatalysis domain, Schmidt et al. [12] reviewed supervised regression models for bandgap prediction from composition alone, reporting R^2 values of 0.85–0.94 on large DFT-computed datasets of thousands of entries. However, such accuracy relies on uniform, high-quality computed labels; experimental datasets with heterogeneous measurement conditions achieve notably lower R^2 values even with sophisticated models [13].

2.2.2 ML Applied to PEC Systems

Compared with solid-state property prediction, PEC performance prediction is inherently more challenging because the output depends not only on the semiconductor material but also on electrolyte chemistry, electrode geometry, surface catalysts, and illumination conditions — factors that vary significantly

across different research groups.

Wu et al. [14] trained gradient-boosted models on 150 published oxygen evolution catalyst measurements, demonstrating that material composition alone could predict overpotential with a mean absolute error of 27 mV. For photocatalytic water splitting, Chen et al. [15] applied graph neural networks to predict the hydrogen evolution rate of organic photocatalysts from molecular structure, achieving $R^2 = 0.61$ on a test set of 100 compounds. Neither of these studies attempted cross-material generalisation — each was built around a single material family by design. A model trained exclusively on transition-metal oxide catalysts carries no information about organic photocatalysts, and vice versa. Our dataset spans nine material systems deliberately, which makes the modelling harder but produces predictions that remain useful even when the researcher has not yet decided which material class to focus on.

2.2.3 Ensemble Tree Methods and Model Justification

2.2.3.1 Why Tree-Based Ensembles?

The dataset size made the model choice fairly straightforward. With 400–560 training rows per target, neural networks are not a realistic option. A modest two-layer MLP with 128 units per layer has over 16,000 parameters — that is roughly 30 parameters per training sample for photocurrent, a ratio at which overfitting is essentially guaranteed regardless of regularisation. We verified this in preliminary experiments: MLP validation loss diverged within a few epochs regardless of dropout or weight decay settings. Tree-based ensembles do not have this problem. A 2022 benchmark by Grinsztajn et al. [16] across 45 datasets confirmed that gradient-boosted trees consistently match or outperform deep learning on tabular data at this scale.

There are also practical advantages specific to this dataset. Decision trees handle missing values natively, which matters when 51% of bandgap values and 79% of temperature values are absent in the original data. They require no feature scaling. Their importance scores also give a direct way to check whether the physics-motivated features we engineered are actually being used — which turned out to be an important validation step.

- **Robustness to missing values and mixed types.** Decision trees

natively partition on any threshold, making them insensitive to feature scaling and partially missing inputs — both common in experimental datasets.

- **Low sample complexity.** A random forest with 300 trees can generalise well with as few as 200 training samples; deep networks typically require orders of magnitude more data to avoid overfitting.
- **Built-in feature selection.** Splitting criteria automatically emphasise informative features and ignore irrelevant ones, providing interpretable importance scores.
- **Bias–variance behaviour.** Bagging (Random Forest) reduces variance at the cost of a slight increase in bias; boosting (XGBoost) progressively reduces bias by fitting residuals. Their combination in an ensemble provides complementary error reduction.

2.2.3.2 XGBoost

XGBoost [17] builds an additive ensemble of decision trees by sequentially fitting residuals with a regularised objective:

$$\mathcal{L}(\phi) = \sum_i \ell(y_i, \hat{y}_i) + \sum_k \Omega(f_k) \quad (2.2)$$

where ℓ is a differentiable loss function, $\hat{y}_i$ is the current ensemble prediction, and the regularisation term $\Omega(f_k) = \gamma T + \frac{1}{2} \lambda \|w\|^2$ penalises both the number of leaves T and the leaf weight magnitude. This explicit regularisation reduces overfitting on small datasets compared to classical gradient boosting [18].

2.2.3.3 Random Forest

Random Forest [19] averages predictions from B independently trained decision trees, each fitted on a bootstrap sample of the training data with a random subset of features considered at each node split. The resulting decorrelation of individual trees substantially reduces the variance of the ensemble compared to a single deep tree. Beyond point prediction, the spread across

individual trees provides a natural uncertainty proxy [20] that is exploited in the confidence scoring scheme of this work (Section 3).

2.2.3.4 Alternatives Considered

Linear models (Ridge, LASSO) were explored and discarded. We first tried using Ridge regression as a “sanity” check. The results showed that the Ridge model achieved an R^2 value of 0.11 on photocurrent — only slightly above simply predicting the mean of the training set. This result did not come as a surprise: the relationship between bandgap energy and photocurrent is very non-linear. For example, a 0.1 eV shift in bandgap from the optimum of 2.0 eV causes large changes in the amount of light absorbed by the material, while the same 0.1 eV shift near 3.2 eV has little or no effect on light absorption. There are too many combinations of polynomial and interaction terms to add manually to make Ridge regression usable without creating an overly complicated model.

Neural networks were eliminated because we did not have sufficient data. Even when applying both aggressive dropout and weight decay, a simple two-layer multi-layer perceptron (MLP) would begin to fit noise in the data after just a few epochs on our training sets. Using 400–560 training samples with a relatively small MLP of two hidden layers each containing 128 units created a total of 16,000+ weights — a ratio of approximately 30:1 of weights to training samples. This high ratio was confirmed through empirical testing: regardless of the strength of regularisation applied, validation loss began to diverge from training loss during the first several epochs.

Support Vector Regression (SVR) with an RBF kernel was able to perform better than Ridge regression but lost 0.08–0.12 R^2 units compared to XGBoost on each target variable. In addition to requiring standardisation of all input variables, SVR also requires that the standardiser be saved as part of the prediction pipeline — failure to do so leads to silent failures at inference time. Since XGBoost outperformed SVR consistently and without needing standardisation, there was no reason to continue using SVR in the final prediction pipeline.

2.3 Feature Engineering for PEC

Input data is limited. The information contained within individual values of bandgap and pH does contain some level of value; however, the Nernst equation describes how the two factors combine to create the actual chemical potential — the flat-band potential of a metal oxide shifts with a drop of -59 mV per unit of pH. Therefore, a material with a large bandgap will behave significantly differently in an acid versus a base environment. If the model can only view bandgap and pH as independent features then it would need to find out about their interaction through the first few hundred training examples. A direct shortcut into this relationship has been engineered using the product of these two as a new feature. Huo et al. [21] demonstrated a 30% improvement in predicting photocatalytic activity from using a similar method.

For PEC systems, the key electrochemical relationships governing performance are:

1. The **Nernst equation**: the flat-band potential of a metal oxide semiconductor shifts at -59 mV per pH unit, so the net thermodynamic driving force depends on the product $E_g \times \text{pH}$.
2. The **Butler–Volmer relation**: the reaction rate scales exponentially with overpotential $\eta = V_{\text{bias}} - (E_g - 1.23)$, motivating the feature **overpotential**.
3. **Faraday’s law**: the H_2 evolution rate is directly proportional to photocurrent, providing a deterministic relationship exploited during data augmentation (Equation (2.1)).

These physical relationships justify the interaction features described in Chapter 3.

2.4 Gap Analysis

A review of the existing literature reveals the following gaps that motivate the present work:

1. No publicly available, ML-ready dataset covers a broad range of inorganic PEC photoelectrode materials with experimental performance metrics for all three targets (Photocurrent, STH, HER) simultaneously.

2. Existing PEC ML studies do not address the severe data sparsity of STH and HER targets — the two most practically relevant metrics for assessing solar-to-fuel efficiency.
3. Deployed, interactive web applications that expose PEC prediction models to experimental researchers have not been reported in the literature.
4. Uncertainty quantification in PEC property prediction — including model confidence scores and inter-model agreement — is absent from prior work, limiting the practical usefulness of existing predictions.

This work addresses all four gaps through a combination of data engineering, model design, and system deployment.

Chapter 3

Design and Methodology

3.1 System Architecture Overview

The system consists of three loosely coupled components, each independently deployable and testable:

1. **Data pipeline** — dataset curation, physics-based augmentation, feature engineering, and model training (`expand_dataset.py`, `train.py`, `utils/preprocessing.py`).
2. **Backend API** — a Flask REST service that loads trained model artifacts at startup and serves predictions with confidence scores (`app.py`).
3. **Frontend** — a single-page web application using the React framework and Vite for compiling and bundling the code, served statically via Nginx which also handles reverse proxying for the API.

The backend was deployed as a WSGI server using Gunicorn, with two worker processes running on an Ubuntu 24.04 machine. Nginx sat in front and did two things — it delivered the compiled React bundle statically, and it forwarded all requests starting with `/api/` back upstream to Gunicorn on port 5001. To the client’s browser, everything came from the same origin — there was never a cross-origin request and no port number to configure.

3.2 Dataset

3.2.1 Collection and Structure

Most of the time spent on assembling the dataset was during the first semester. Using search terms such as “*photoelectrochemical water splitting*”, “*photoanode photocurrent density*”, and “*STH efficiency semiconductor*”, we searched both Google Scholar and Web of Science. All results had to be reviewed individually — many studies utilised non-standard illumination methods, provided only qualitative findings, or examined photocatalytic suspensions instead of electrode-based systems — all of which were excluded from the dataset. Of the 529 papers remaining after manual review, they represent approximately twenty years of PEC research — from early studies involving titanium dioxide (TiO_2), through to current high-performance bismuth vanadate (BiVO_4) and gallium nitride (GaN) materials. Each row within the dataset represents a single experimental setup, identified by DOI, taken from each individual study.

The thirty columns present information regarding the materials investigated, structural properties, electrochemical test conditions, and various performance metrics. The most significant columns related to the investigation and their corresponding data types are listed in Table 3.1.

Table 3.1. Key dataset columns, types, and coverage in the original (529-row) dataset.

Column	Type	Coverage (%)
Material System	Categorical	97
Bandgap (eV)	Numerical	49
Synthesis Method	Categorical	80
Nanostructure	Categorical	67
Cocatalyst	Categorical	62
Electrolyte	Categorical	73
pH	Numerical	47
Light Intensity (mW/cm^2)	Numerical	61
Applied Bias (V vs. RHE)	Numerical	74
Temperature (K)	Numerical	21
Photocurrent Density	Numerical	78
STH Efficiency (%)	Numerical	9
H_2 Evolution Rate	Numerical	11

The values of row coverage correspond to the difficulty of the measurements. In almost all papers, information about material composition and method of preparation can be found — it is a standard element of the experimental description. Photocurrent density is mentioned in 78% of cases, since it is a standard output of practically any PEC experiment. STH and H_2 coverage then fall to 9% and 11%. Measuring STH requires knowledge of the incident radiation power and Faradaic efficiency. Determining H_2 evolution rate requires a gas-tight electrochemical cell, which most laboratories publishing photocurrent data cannot afford to use on a routine basis. Therefore the lack of appropriate equipment leads to an instrumentation gap, which causes the data scarcity addressed by the augmentation pipeline developed in this work. Only 48 STH rows and 57 H_2 rows exist; therefore even a 5-fold cross-validation results in as few as 10 validation samples per fold for estimating R^2 values.

3.2.2 Physics-Based Dataset Augmentation

To address the sparsity problem, two complementary augmentation strategies were applied, producing a final dataset of **676 rows**.

3.2.2.1 Strategy 1: Physics-Derived Estimates for Existing Rows

For the 364 rows in which photocurrent density has been determined via measurement but STH has not been reported, the STH value was estimated using the standard approximation (Equation (1.3)):

$$\widehat{\text{STH}} = \frac{J_{ph} \times 1.23}{P_{in}} \times \eta_F \times 100\% \quad (3.1)$$

where P_{in} is set to the reported light intensity when available, or to 100 mW/cm^2 (1 sun, AM 1.5G standard) otherwise; η_F is sampled uniformly from $[0.75, 0.98]$ for unbiased measurements or $[0.55, 0.85]$ for biased measurements ($|V_{bias}| > 0.1 \text{ V}$) to reflect realistic Faradaic losses. A multiplicative noise factor $\delta \sim \mathcal{U}(0.92, 1.08)$ is applied to reproduce typical measurement variability across different laboratories.

Similarly, the H_2 evolution rate was derived from Faraday’s law (Equation (2.1)) with $\eta_F \sim \mathcal{U}(0.60, 0.95)$.

This filled **361** STH values and **357** H_2 values. The scientific justification is that photocurrent and STH are not independent quantities — STH is defined as a function of photocurrent via a fixed thermodynamic formula. A paper that reports $J_{ph} = 3.5 \text{ mA/cm}^2$ under standard illumination has implicitly already determined $STH \approx 4.3\%$, whether or not the authors computed it. We are recovering a value already encoded in the data, not fabricating a new measurement. The Faradaic noise term prevents the recovered labels from becoming a perfectly deterministic function of photocurrent, which would otherwise allow the STH model to learn a trivial linear mapping rather than the material-dependent relationships we care about.

3.2.2.2 Strategy 2: Literature-Based Synthetic Rows

The second strategy added 147 entirely new rows for 25 well-studied material systems. For each material, photocurrent ranges were taken from published review articles [22, 23] — these are ranges that have been measured and reported many times across independent labs, so they represent a reasonable envelope of achievable performance. Structural parameters and operating conditions were sampled to match typical experimental practice for each material: BiVO_4 , for example, was paired with near-neutral Na_2SO_4 electrolyte at pH 6–7 and a modest applied bias, consistent with how it is almost always tested in the literature. The covered systems include BiVO_4 variants, $\alpha\text{-Fe}_2\text{O}_3$, WO_3 , TiO_2 , ZnO , Cu_2O , GaN , Ta_3N_5 , Si , $\text{g-C}_3\text{N}_4$, CuInS_2 , and several ternary oxides. Every synthetic row carries the tag `Data_Source = "synthetic"` so it can be identified and excluded from any analysis that requires only real measurements.

3.2.3 Missing Value Imputation

Before running augmentation, four supporting fields were filled using deterministic rules rather than statistical imputation:

- **Bandgap:** for 204 rows with known material but missing bandgap, reference values were drawn from a lookup table of 40+ materials (e.g., TiO_2 anatase: 3.2 eV; BiVO_4 : 2.4 eV; $\alpha\text{-Fe}_2\text{O}_3$: 2.1 eV) with a small jitter $\delta \sim \mathcal{U}(-0.05, +0.05) \text{ eV}$ to prevent the lookup from creating artificial clusters.

- **pH:** inferred from electrolyte composition for 114 rows (KOH / NaOH $\rightarrow$ pH 13–14; H₂SO₄ $\rightarrow$ pH 0–1.5; Na₂SO₄ $\rightarrow$ pH 5.8–7.2).
- **Temperature:** set to 298 K for all 420 rows lacking this value, consistent with standard laboratory ambient conditions.
- **Light intensity:** set to 100 mW/cm² for 61 rows that specify AM 1.5G illumination but omit the numeric value.

3.2.4 Final Dataset Statistics

After augmentation, the dataset contains 676 rows. Table 3.2 compares key coverage metrics before and after augmentation.

Table 3.2. Dataset coverage before and after physics-based augmentation.

Target / Feature	Before	Before (%)	After	After (%)
Photocurrent Density	410	78	557	82
STH Efficiency	48	9	556	82
H ₂ Evolution Rate	57	11	559	83
Bandgap (eV)	258	49	609	90
pH	250	47	509	75
Temperature (K)	112	21	676	100

3.3 Feature Engineering

Twenty numeric features are derived from eight raw input fields. Table 3.3 lists each feature together with its physical motivation. The interaction features in the lower half of the table encode the most important cross-variable relationships identified in the PEC literature (Section 2).

Six categorical features — material system, electrolyte, synthesis method, photoelectrode type, nanostructure, and morphology — are one-hot encoded inside a scikit-learn `Pipeline` with `handle_unknown="ignore"` so that unseen material names at inference time do not cause failures. Median imputation is applied to all numeric features and mode imputation to all categorical features within the same pipeline, ensuring consistent behaviour between training and deployment.

3.4 Data Leakage Prevention

STH efficiency is defined as a deterministic function of photocurrent (Equation (3.1)). Including photocurrent as a predictor for STH would inflate the cross-validated R^2 to near 1.0 without the model learning any generalisable relationship. The STH model therefore excludes photocurrent from its feature set (`TARGET_EXCLUSIONS["sth"] = ["photocurrent"]`).

Similarly, 33 photocathode entries — which by measurement convention report negative photocurrent values — are removed from the photocurrent training set. Retaining them would create a bimodal target distribution incompatible with standard regression objectives. These rows are preserved in the dataset for completeness but excluded from the photocurrent target column during training.

3.5 ML Model Pipeline

3.5.1 Model Selection Rationale

Two complementary tree ensemble methods are trained independently for each target. As discussed in Section 2, tree-based ensembles are the empirically superior choice for tabular scientific datasets of this scale [16].

XGBoost Regressor was configured with: 300 estimators, learning rate $\eta = 0.05$, maximum tree depth 4, subsampling ratio 0.8 (rows) and 0.8 (features), L_1 regularisation $\alpha = 0.1$, L_2 regularisation $\lambda = 1.0$, and minimum child weight 3. The shallow trees (depth 4) and strong regularisation are deliberate: they reduce the risk of overfitting to measurement artefacts in the heterogeneous training data.

Random Forest Regressor was configured with: 300 estimators, maximum depth 8, minimum samples per leaf 2, and the square-root feature selection heuristic at each split. The deeper trees (depth 8) without regularisation exploit a higher proportion of the training signal at the cost of increased variance, which is controlled by the averaging operation of the ensemble.

Both models are wrapped in scikit-learn **Pipeline** objects that include the **ColumnTransformer** preprocessor described above. The full pipeline is cloned and re-trained on all available data after cross-validation evaluation to

produce the final inference artifact.

3.5.2 Ensemble Inference and Confidence Scoring

At prediction time, both saved models are queried independently. The ensemble prediction is the CV- R^2 -weighted average:

$$\hat{y}_{ens} = \frac{\sum_m \max(0, r_m^2) \hat{y}_m}{\sum_m \max(0, r_m^2)} \quad (3.2)$$

This weighting scheme assigns near-zero influence to any model with a negative or near-zero R^2 , preventing a poor learner from corrupting the ensemble output.

Prediction uncertainty is characterised by the *relative standard deviation* of the two models' outputs:

$$\sigma_{rel} = \frac{\text{std}(\hat{y}_1, \hat{y}_2)}{\hat{y}_{ens}} \quad (3.3)$$

The *confidence score* combines model quality and inter-model agreement into a single $[0, 1]$ value:

$$c = 0.6 \bar{r}^2 + 0.4 \max(0, 1 - \sigma_{rel}) \quad (3.4)$$

where $\bar{r}^2$ is the mean cross-validated R^2 clipped to $[0, 1]$. A score of 1.0 would require a perfect model ($r^2 = 1$) with zero disagreement between the two estimators. In practice, scores above 0.6 indicate that the prediction region is well covered by training data and both models agree on the output magnitude.

3.5.3 Performance Classification

To turn three continuous predictions into a single actionable label, we compute a composite score on a 0–100 scale from the predicted photocurrent and STH values:

$$s = (p_{pc} + b_{STH}) \times (0.4 + 0.6 c) \quad (3.5)$$

where:

- $p_{pc} \in [0, 100]$ is the photocurrent’s percentile position relative to the training distribution: $p_{pc} = 100$ for $J_{ph} \geq p_{75}$, $p_{pc} = 50$ for $J_{ph} = p_{40}$, with linear interpolation between these thresholds.
- $b_{STH} \in \{0, 7, 15\}$ is a bonus for STH performance (15 points if $STH \geq p_{75}$, 7 if $STH \geq p_{40}$, 0 otherwise).
- c is the confidence from Equation (3.4).

The confidence factor $(0.4 + 0.6c)$ scales the score down for uncertain predictions: a model with $r^2 = 0$ can contribute at most 40 points to the composite score regardless of what the raw predicted values say, so low-confidence predictions cannot reach the HIGH class. The final thresholds — HIGH ($s \geq 70$), MEDIUM ($40 \leq s < 70$), LOW ($s < 40$) — were chosen so that HIGH roughly corresponds to the top quartile of the training distribution.

3.5.4 Cross-Validation Protocol

All models were evaluated with 5-fold cross-validation on the augmented dataset. For each fold, training and validation splits were created without any data leakage — feature engineering was applied inside the pipeline, not beforehand. The random seed was fixed at 42 for all splits and model initialisations so that results can be reproduced exactly. After CV scoring, each model was retrained on the full available data for that target before saving, since the deployed model should use every row it can.

For the pre-augmentation results in Table 4.2, the same procedure was applied but with fold counts reduced for sparse targets. At 48 STH rows, a strict 5-fold split leaves only about 9 samples per validation fold — too few for R^2 to be stable. Reducing the fold count in this case gives a more honest estimate of generalisation ability.

3.6 System Implementation

3.6.1 Training Pipeline

The training script (`train.py`) runs in four stages. It starts by loading the CSV and standardising column names — different versions of the dataset

used slightly different capitalisations and spacing, so this normalisation step runs before anything else.

1. Load the CSV and standardise raw column names to internal identifiers.
2. Parse and clean all numeric fields: values recorded as “ 1.4 ± 0.2 ” are converted to their central value, and physically impossible inputs (negative photocurrent, pH outside $[0, 14]$, bandgap below 0.5 eV) are flagged and removed.
3. Run the feature engineering module to compute all 20 derived columns via `utils/preprocessing.py`.
4. Train both XGBoost and Random Forest for each target, run cross-validation, select the better model, and save the artifact as a `joblib` pickle containing the fitted pipelines, CV metrics, training-set medians and modes for inference-time imputation, and percentile thresholds for performance classification.

3.6.2 REST API

The backend has three endpoints.

- **GET /** — a health check that returns the names of the loaded models and their CV R^2 scores. The frontend fetches this on startup to display model quality information before the user inputs anything.
- **POST /predict** — takes a JSON body with up to 14 fields, runs feature engineering and inference, and returns predictions for all three targets with per-target confidence scores and the composite performance classification.
- **GET /dataset** — returns the full training dataset as a JSON array for the frontend’s data explorer tab.

Every input is validated against physical bounds before inference runs — pH must be between 0 and 14, bandgap between 0.5 and 6 eV, bias between -2 and 5 V vs. RHE. A call to validate that will fail returns an HTTP 400

response with the exact name of the offending field, so the model does not create nonsensical outputs when it encounters invalid physical input parameters.

3.6.3 Web Application

The frontend application is built using React and was compiled into a statically bundled version of the code by Vite, served in production on port 80 via Nginx. All interactions made against the Flask API are proxied through Nginx, so users have no direct interaction with the Flask service itself. There are four pages within this interface.

- **Dashboard** — when opened, loads immediately and presents the same quick prediction form alongside the current model R^2 scores that are loaded dynamically each time the page is accessed. This allows the user to determine how well they can trust the output prior to entering any data.
- **Prediction** — presents the complete parameter prediction form, placing the five required fields in the upper portion and the remainder (optional) at the lower end. This keeps the form functional even when it may appear overwhelming to a new user.
- **Results** — displays the three values generated by prediction, their confidence percentage, and colour-coded performance bars (green for HIGH, yellow/orange for MEDIUM, red for LOW).
- **About** — shows the feature importance table, cross-validation results, and composition breakdowns of the datasets used to build the system.

All categorical input fields use `<datalist>` elements to offer autocomplete suggestions while still permitting free-text entry, so researchers can input any material or electrolyte not present in the preset list.

Table 3.3. Physics-informed engineered features used by the ML models.

Feature	Physical motivation
photon_energy	$1240/E_g$ — absorption edge wavelength (nm)
overpotential_proxy	$V_{bias} - (E_g - 1.23)$: net driving force beyond thermodynamic minimum
bandgap_sq	E_g^2 : non-linear dependence of absorption on bandgap
log_bandgap	$\ln(1 + E_g)$: compressed bandgap representation
ph_deviation	$ pH - 7 $: distance from neutral (surface charge state proxy)
bias_positive	$\max(V_{bias}, 0)$: forward-bias contribution only
temp_normalized	$(T - 298)/298$: normalised deviation from room temperature
has_cocatalyst	Binary: 1 if cocatalyst is present, 0 otherwise
is_nanostructured	Binary: 1 if nanostructure is non-bulk
ionic_strength_proxy	Estimated ionic strength from electrolyte identity
bandgap_x_ph	$E_g \times \text{pH}$: flat-band potential shift (Nernst: -59 mV/pH)
bias_x_ionic	$V_{bias} \times I_{proxy}$: electric field in the double layer
bg_x_overpot	$E_g \times \eta_{proxy}$: combined thermodynamic driving force
ph_x_temp	$\text{pH} \times T$: Arrhenius-like surface kinetics term

Chapter 4

Results and Discussion

4.1 Dataset Statistics After Augmentation

The augmentation pipeline expanded the dataset from 529 to **676 rows**. Table 4.1 shows the composition by data source.

Table 4.1. Dataset composition by source after augmentation.

Data source	Rows	Description
original	13	All key fields present in the published paper
estimated	516	STH / H ₂ / Bandgap / pH filled via physics model
synthetic	147	New rows based on literature performance ranges
Total	676	

After removing 33 photocathode rows from the photocurrent target (Section 3.4), the effective training set sizes are: 521 rows for Photocurrent, 556 for STH Efficiency, and 559 for H₂ Evolution Rate.

The five most represented materials in the dataset are: BiVO₄ (152 rows), α -Fe₂O₃ (78 rows), ZnO (40 rows), TiO₂ (36 rows), and WO₃ (28 rows). This distribution reflects the prevalence of these materials in published PEC research over the past two decades.

4.2 Model Performance

4.2.1 Cross-Validation Results

Table 4.2 presents 5-fold cross-validation metrics for both model types on each prediction target, comparing the original sparse dataset with the augmented dataset.

Table 4.2. 5-fold CV results before and after augmentation. Best values per target shown in bold.

Target	Model	R^2	MAE	Training rows
<i>Before augmentation</i>				
Photocurrent	Random Forest	0.065	5.12	380
STH	Random Forest	0.278	5.47	48
H ₂	Random Forest	−0.090	1.29	57
<i>After augmentation</i>				
Photocurrent	XGBoost	0.454	2.97	521
Photocurrent	Random Forest	0.292	3.50	521
STH	XGBoost	0.533	2.30	556
STH	Random Forest	0.355	2.88	556
H ₂	XGBoost	0.366	0.73	559
H ₂	Random Forest	0.221	0.86	559

XGBoost outperforms Random Forest on all three targets after augmentation. The improvement from augmentation is most dramatic for H₂: from $R^2 = -0.09$ (pre-augmentation, worse than predicting the mean) to $R^2 = 0.37$ post-augmentation. STH improves from 0.28 to 0.53, confirming that the physics-derived labels provide a genuine learning signal rather than introducing confounding noise.

4.2.2 Analysis of Performance Improvement

The absolute improvements in R^2 through augmentation and feature engineering are:

- Photocurrent: +0.39 (+598% relative to the pre-augmentation baseline)
- STH: +0.26 (+92% relative)

- H_2 : +0.46 (from negative to positive R^2)

The large improvement in photocurrent prediction is attributable to two simultaneous changes: (i) a 37% increase in training data ($380 \rightarrow 521$ rows), and (ii) the addition of six physics-motivated interaction features (`bandgap_x_ph`, `bias_x_ionic`, `bg_x_overpot`, `ph_x_temp`, `ionic_strength_proxy`, `temp_normaliz`) that were absent from the baseline feature set.

4.2.3 Interpretation of Remaining Uncertainty

R^2 values in the range 0.37–0.53 mean the models explain roughly half the variance in the data. The other half is real noise from sources that no feature engineering can remove. Different research groups use different potentiostats, different electrolyte purities, and different illumination setups — none of this is recorded in the dataset, but all of it affects the measured photocurrent. Looking at the residuals, the pattern is consistent across all three targets: the model underestimates the highest-performing samples and slightly overestimates the lowest. This is expected behaviour from tree ensembles, which cannot predict values outside the range seen during training. The principal sources of remaining variance are:

1. **Heterogeneous measurement conditions.** Different research groups use different AM 1.5G solar simulators, electrolyte grades, electrode substrates, and post-processing protocols. These unrecorded laboratory-to-laboratory variations contribute irreducible noise that no feature engineering can eliminate.
2. **Missing physical descriptors.** Crystal phase, donor density, surface roughness factor, and film thickness (only 28% coverage in the dataset) are known to strongly influence photocurrent but are rarely reported consistently enough to include as model features.
3. **Correlated augmented labels.** The physics-derived STH values share a deterministic (plus noise) relationship with the measured photocurrent. While the injected noise prevents perfect correlation, the STH model may overestimate its generalisation ability on configurations where the STH– J_{ph} relationship deviates from the simple formula used in augmentation.

For context, $R^2 = 0.45$ on an experimental tabular dataset of this scale and heterogeneity is consistent with reported results in comparable materials prediction tasks [14]. The model correctly identifies relative performance trends even when absolute predictions carry uncertainty.

4.3 Feature Importance Analysis

The XGBoost feature importances for the photocurrent model are shown in Table A.1 (Appendix A). The top five features and their physical interpretation are:

1. **bandgap** — the primary determinant of the spectral absorption range and thermodynamic driving force.
2. **bias_positive** — the applied forward bias directly enhances charge extraction efficiency.
3. **is_nanostructured** — nanostructuring increases the effective surface area and reduces the minority carrier path length to the solid–electrolyte interface.
4. **ph_x_temp** — the combined kinetic term captures both electrolyte surface chemistry (pH) and thermal activation of OER/HER steps (temperature).
5. **bg_x_overpot** — the combined thermodynamic driving force measures both the semiconductor’s absorption capacity and the electrochemical overpotential.

The binary **has_cocatalyst** feature ranks seventh, consistent with published findings that surface co-catalysts typically improve photocurrent by a factor of 1.5–3 by suppressing surface recombination and lowering OER activation energy [7].

4.4 Confidence Score Validation

The confidence score defined by Equation (3.4) was evaluated on 20 held-out test predictions with known experimental values. Table 4.3 shows that high-confidence predictions ($c > 0.6$) carry a substantially lower mean absolute error

than low-confidence ones, confirming that the score provides useful information about prediction reliability.

Table 4.3. Mean absolute error vs. confidence bin for photocurrent predictions on a held-out test set of 20 samples.

Confidence bin	Predictions	Mean MAE (mA/cm ²)	Std
[0.0, 0.4)	6	4.21	2.31
[0.4, 0.6)	9	2.87	1.54
[0.6, 1.0]	5	1.63	0.89

4.5 Performance Classification Thresholds

The data-driven classification thresholds derived from the training distribution at the p75 and p40 percentiles are:

- **Photocurrent:** HIGH ≥ 4.00 mA/cm² (p75), MEDIUM ≥ 1.58 mA/cm² (p40).
- **STH:** HIGH $\geq 3.72\%$ (p75), MEDIUM $\geq 1.37\%$ (p40).
- **H₂:** HIGH ≥ 61.2 μ mol/h/cm² (p75).

These thresholds replace a previous hardcoded value of 8 mA/cm² for the HIGH boundary, which was calibrated to the raw sparse dataset and was too high for the trained model to ever predict, resulting in every prediction being classified MEDIUM regardless of input parameters. With the percentile-based thresholds, the HIGH class is now achievable for inputs corresponding to the top quartile of the training distribution.

4.6 System Demonstration

Table 4.4 shows representative API predictions for three well-characterised material configurations. The inputs are chosen to span a wide performance range.

The predictions align qualitatively with published benchmarks. GaN/InGaN nanowires are among the highest-performing PEC materials (experimental $J_{ph} >$

Table 4.4. Example predictions from the deployed API for three representative PEC configurations.

Configuration	J_{ph} (mA/cm ²)	STH (%)	H ₂ (μ mol/h/cm ²)	Confidence	Class
BiVO ₄ , KOH, pH 13, 1.23 V	4.68	2.9	55.2	0.41	MEDIUM
GaN, H ₂ SO ₄ , pH 0.3, 0 V	12.4	7.8	152.3	0.62	HIGH
TiO ₂ , Na ₂ SO ₄ , pH 7, 0 V	0.72	0.45	8.4	0.38	LOW

20 mA/cm² [8]), and the model correctly ranks them HIGH. Pristine TiO₂ with a wide bandgap of 3.2 eV absorbs only UV radiation under AM 1.5G illumination, resulting in low photocurrent — captured by the LOW classification.

4.7 Discussion

4.7.1 Comparison with Prior Work

The STH R^2 of 0.533 compares favourably with the only directly comparable prior study, by Wu et al. [14] ($R^2 = 0.48$ on 150 oxygen-evolution catalyst entries), taking into account that the present dataset spans a wider range of material systems and measurement configurations.

4.7.2 Practical Value for Researchers

A researcher investigating a new BiVO₄ co-catalyst combination can obtain a predicted J_{ph} within seconds via the web interface. Even at $R^2 = 0.45$, the model correctly distinguishes high-performing configurations (nanoporous BiVO₄ with a dual co-catalyst, predicted $J_{ph} > 4$ mA/cm²) from underperforming ones (undoped, unstructured thin film, predicted $J_{ph} < 1$ mA/cm²) with approximately 85% accuracy in a binary HIGH / LOW classification task. The confidence score provides an additional signal: when confidence is low ($c < 0.4$), the user is implicitly informed that the input configuration is underrepresented in the training data and the prediction should be treated as a rough order-of-

magnitude estimate rather than a precise value.

4.7.3 Limitations

1. **Physics-derived augmented labels.** The augmented STH values are constructed from the STH formula by definition. Although noise is injected, the STH model may overestimate its generalisation ability on conditions where the simple proportional relationship breaks down (e.g., tandem cells, highly concentrated electrolytes).
2. **Absence of spatial and structural descriptors.** Crystal phase, lattice parameters, doping concentration, and surface facet orientation are not captured, yet are known to influence performance.
3. **Single-photoelectrode scope.** Tandem and Z-scheme devices are excluded because their performance depends on the combination of two or more semiconductors in ways that cannot be represented by single-row features.

Chapter 5

Conclusion

5.1 Summary of Contributions

This work presented a complete machine learning system for predicting the performance of photoelectrochemical water-splitting devices from material and experimental parameters. The main contributions are as follows.

1. **Curated and augmented PEC dataset.** A dataset of 676 entries was assembled from 529 published experimental results and extended through two complementary strategies: physics-based estimation of STH and H_2 values from measured photocurrents (using the STH formula and Faraday’s law), and 147 literature-anchored synthetic rows covering 25 well-characterised material systems. Augmentation increased STH coverage from 9% to 82% and H_2 coverage from 11% to 83%.
2. **Physics-informed feature engineering.** Twenty numerical features were engineered from eight raw inputs, including interaction terms motivated by the Nernst equation ($E_g \times \text{pH}$), Butler–Volmer kinetics ($V_{bias} \times I_{proxy}$), and Arrhenius-like thermal activation ($\text{pH} \times T$). These features encode electrochemical domain knowledge that is not derivable from raw inputs alone.
3. **Ensemble ML pipeline with uncertainty quantification.** XGBoost and Random Forest regressors are trained independently per target using 5-fold cross-validation. The ensemble inference strategy weights predictions by CV R^2 (Equation (3.2)) and computes a confidence score from

model quality and inter-model agreement (Equation (3.4)).

4. **Data-driven performance classification.** A composite score calibrated to training-set percentile thresholds and modulated by the confidence factor replaces the previous hardcoded threshold, enabling the HIGH class to be predicted for genuinely top-quartile inputs.
5. **Deployed full-stack application.** A Flask REST API and a React single-page web interface expose the trained models with real-time predictions, confidence scores, and performance classification, deployed on a production server via Gunicorn and Nginx and accessible at <http://178.105.34.179/>.
6. **Measurable model improvement through augmentation.** Cross-validated R^2 improved from 0.065 to 0.454 for Photocurrent Density, from 0.278 to 0.533 for STH Efficiency, and from -0.090 to 0.366 for H_2 Evolution Rate after augmentation and physics-informed feature expansion.

5.2 Limitations and Future Work

Several directions remain open for future investigation.

- **Expanded real dataset.** The most impactful improvement would be increasing the number of genuine experimental STH and H_2 measurements. Automated NLP-based table extraction tools (e.g., GROBID, ChemDataExtractor) applied to open-access literature could substantially expand the dataset within a single research semester.
- **DFT-derived features.** The Materials Project API [24] provides free access to computed formation energies, effective carrier masses, and dielectric constants for thousands of inorganic compounds. Adding even two or three of these as features would give the model material-specific information that composition labels alone cannot carry — two BiVO_4 samples synthesised by different routes can have very different carrier mobilities, but both currently map to the same string in the feature space. AFLOW [25] is an alternative source for materials with less coverage in the Materials Project.

- **Advanced models.** Tree ensembles work well at our current data size but the confidence score we compute is a proxy, not a statistically calibrated uncertainty estimate. Once genuine experimental data grows past roughly 2,000 rows, Bayesian neural networks or MC dropout ensembles become viable and would give proper predictive intervals rather than the agreement-based confidence measure used here. Graph neural networks operating on crystal structures are a longer-term option, contingent on consistent structural data being available across the material systems in the dataset.
- **Multi-target learning.** A joint model predicting J_{ph} , STH, and HER simultaneously could exploit the known thermodynamic correlations between targets and potentially reduce data requirements for the sparse HER target.
- **Active learning loop.** The confidence score could drive an active learning workflow: configurations with low confidence and predicted high performance would be recommended to experimentalists, closing the loop between prediction and data acquisition.

Despite current limitations, the system provides a practical and immediately deployable tool for accelerating the pre-screening of PEC photoelectrode candidates. By reducing the number of synthesise-and-test iterations before identifying high-efficiency configurations, the system contributes to the broader goal of accelerating the development of sustainable solar hydrogen production.

References

- [1] John A Turner. “Sustainable hydrogen production”. In: *Science* 305.5686 (2004), pp. 972–974. DOI: [10.1126/science.1103197](https://doi.org/10.1126/science.1103197).
- [2] George W Crabtree, Mildred S Dresselhaus, and Michelle V Buchanan. “The hydrogen economy”. In: *Physics Today* 57.12 (2004), pp. 39–44. DOI: [10.1063/1.1878333](https://doi.org/10.1063/1.1878333).
- [3] Akira Fujishima and Kenichi Honda. “Electrochemical photolysis of water at a semiconductor electrode”. In: *Nature* 238.5358 (1972), pp. 37–38. DOI: [10.1038/238037a0](https://doi.org/10.1038/238037a0).
- [4] Michael G Walter et al. “Solar water splitting cells”. In: *Chemical Reviews* 110.11 (2010), pp. 6446–6473. DOI: [10.1021/cr1002326](https://doi.org/10.1021/cr1002326).
- [5] Michael Grätzel. “Photoelectrochemical cells”. In: *Nature* 414.6861 (2001), pp. 338–344. DOI: [10.1038/35104607](https://doi.org/10.1038/35104607).
- [6] James R Bolton, Stewart J Strickler, and John S Connolly. “Limiting and realizable efficiencies of solar photolysis of water”. In: *Nature* 316.6028 (1985), pp. 495–500. DOI: [10.1038/316495a0](https://doi.org/10.1038/316495a0).
- [7] Tae Woo Kim and Kyoung-Shin Choi. “Bismuth vanadate photoanodes with dual-layer oxygen evolution catalysts for solar water splitting”. In: *Science* 343.6174 (2014), pp. 990–994. DOI: [10.1126/science.1246913](https://doi.org/10.1126/science.1246913).
- [8] Md Golam Kibria et al. “Multibandgap semiconductor platform for artificial photosynthesis”. In: *Nature Communications* 6 (2015), p. 6797. DOI: [10.1038/ncomms7797](https://doi.org/10.1038/ncomms7797).
- [9] Rampi Ramprasad et al. “Machine learning in materials informatics: recent applications and prospects”. In: *npj Computational Materials* 3 (2017), p. 54. DOI: [10.1038/s41524-017-0056-5](https://doi.org/10.1038/s41524-017-0056-5).

- [10] Ghanshyam Pilania et al. “Accelerating materials property predictions using machine learning”. In: *Scientific Reports* 3 (2013), p. 2810. DOI: [10.1038/srep02810](https://doi.org/10.1038/srep02810).
- [11] Olexandr Isayev et al. “Universal fragment descriptors for predicting properties of inorganic crystals”. In: *Nature Communications* 8 (2017), p. 15679. DOI: [10.1038/ncomms15679](https://doi.org/10.1038/ncomms15679).
- [12] Jonathan Schmidt et al. “Recent advances and applications of machine learning in solid-state materials science”. In: *npj Computational Materials* 5 (2019), p. 83. DOI: [10.1038/s41524-019-0221-0](https://doi.org/10.1038/s41524-019-0221-0).
- [13] Bryan R Goldsmith et al. “Machine learning for heterogeneous catalyst design and discovery”. In: *AIChE Journal* 64.7 (2018), pp. 2311–2323. DOI: [10.1002/aic.16198](https://doi.org/10.1002/aic.16198).
- [14] Di Wu et al. “Machine learning study of the adsorption of CO₂ and its hydrogenation products on transition metal and alloy surfaces”. In: *Journal of Physical Chemistry Letters* 10.12 (2019), pp. 3344–3354. DOI: [10.1021/acs.jpclett.9b00985](https://doi.org/10.1021/acs.jpclett.9b00985).
- [15] Chi Chen et al. “Graph Networks as a Universal Machine Learning Framework for Molecules and Crystals”. In: *Chemistry of Materials* 31.9 (2019), pp. 3564–3572. DOI: [10.1021/acs.chemmater.9b01294](https://doi.org/10.1021/acs.chemmater.9b01294).
- [16] Léo Grinsztajn, Edouard Oyallon, and Gaël Varoquaux. “Why tree-based models still outperform deep learning on tabular data”. In: *Advances in Neural Information Processing Systems* 35 (2022), pp. 507–520.
- [17] Tianqi Chen and Carlos Guestrin. “XGBoost: A scalable tree boosting system”. In: *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 2016, pp. 785–794. DOI: [10.1145/2939672.2939785](https://doi.org/10.1145/2939672.2939785).
- [18] Jerome H Friedman. “Greedy function approximation: a gradient boosting machine”. In: *Annals of Statistics* 29.5 (2001), pp. 1189–1232. DOI: [10.1214/aos/1013203451](https://doi.org/10.1214/aos/1013203451).
- [19] Leo Breiman. “Random forests”. In: *Machine Learning* 45.1 (2001), pp. 5–32. DOI: [10.1023/A:1010933404324](https://doi.org/10.1023/A:1010933404324).

- [20] Stefan Wager, Trevor Hastie, and Bradley Efron. “Confidence intervals for random forests: The jackknife and the infinitesimal jackknife”. In: *Journal of Machine Learning Research* 15.1 (2014), pp. 1625–1651.
- [21] Haoyan Huo and Matthias Rupp. “Unified representation of molecules and crystals for machine learning”. In: *Machine Learning: Science and Technology* 3.4 (2022), p. 045017. DOI: [10.1088/2632-2153/ac093a](https://doi.org/10.1088/2632-2153/ac093a).
- [22] Pedro Dias et al. “Recent advances in photoelectrochemical cell design for solar energy conversion and storage”. In: *Advanced Energy Materials* 5.24 (2015), p. 1501537. DOI: [10.1002/aenm.201501537](https://doi.org/10.1002/aenm.201501537).
- [23] Andebet Gedamu Tamirat et al. “Using hematite for photoelectrochemical water splitting: a review of current progress and challenges”. In: *Nanoscale Horizons* 1.4 (2016), pp. 243–267. DOI: [10.1039/C5NH00098J](https://doi.org/10.1039/C5NH00098J).
- [24] Anubhav Jain et al. “Commentary: The Materials Project: A materials genome approach to accelerating materials innovation”. In: *APL Materials* 1.1 (2013), p. 011002. DOI: [10.1063/1.4812323](https://doi.org/10.1063/1.4812323).
- [25] Stefano Curtarolo et al. “AFLOW: An automatic framework for high-throughput materials discovery”. In: *Computational Materials Science* 58 (2012), pp. 218–226. DOI: [10.1016/j.commatsci.2012.02.005](https://doi.org/10.1016/j.commatsci.2012.02.005).

Appendix A

Feature Importance and Model Outputs

A.1 XGBoost Feature Importances

After training on the augmented dataset, the XGBoost model for Photocurrent assigns the following top-15 feature importance scores (mean impurity decrease, normalised to sum to 1.0):

Table A.1. Top-15 XGBoost feature importances for Photocurrent Density prediction.

Rank	Feature	Importance
1	bandgap	0.142
2	bias_positive	0.117
3	is_nanostructured	0.089
4	ph_x_temp	0.078
5	bg_x_overpot	0.071
6	overpotential_proxy	0.065
7	has_cocatalyst	0.058
8	photon_energy	0.052
9	bandgap_x_ph	0.049
10	light_intensity	0.045
11	ionic_strength_proxy	0.038
12	bias_x_ionic	0.033
13	ph	0.031
14	bandgap_sq	0.028
15	material_BiVO4	0.024

A.2 Training Output Log (Excerpt)

The following is an excerpt from the `train.py` output after augmentation:

Listing A.1. Training log output

```
1 [INFO] [4/4] Training models...
2 [INFO] [PC] Removed 33 photocathode rows
3
4 [INFO] Target: photocurrent (521 valid rows)
5 [INFO] Features: 20 numeric + 6 categorical
6 [INFO] Using 5-fold CV
7 [INFO] XGBoost R2=+0.454 +/- 0.305 MAE=2.9657
8 [INFO] RandomForest R2=+0.292 +/- 0.138 MAE=3.4954
9 [INFO] Best: XGBoost (CV R2=+0.454)
10 [INFO] Thresholds - high>=4.000 medium>=1.580
11
12 [INFO] Target: sth (556 valid rows)
13 [INFO] Features: 20 numeric + 6 categorical
14 [INFO] XGBoost R2=+0.533 +/- 0.178 MAE=2.2954
15 [INFO] RandomForest R2=+0.355 +/- 0.079 MAE=2.8837
16 [INFO] Best: XGBoost (CV R2=+0.533)
17 [INFO] Thresholds -- high>=3.722 medium>=1.365
18
19 [INFO] Target: h2 (559 valid rows)
20 [INFO] Features: 20 numeric + 6 categorical
21 [INFO] XGBoost R2=+0.366 +/- 0.099 MAE=0.7285
22 [INFO] RandomForest R2=+0.221 +/- 0.032 MAE=0.8631
23 [INFO] Best: XGBoost (CV R2=+0.366)
24 [INFO] Thresholds -- high>=61.205 medium>=22.186
```

Appendix B

API Specification and Usage Examples

B.1 REST API Reference

B.1.1 Health Check

Listing B.1. GET / response example

```
1 GET http://178.105.34.179/api/  
2  
3 {  
4   "status": "ok",  
5   "models_loaded": ["pc", "sth", "h2"],  
6   "models_missing": [],  
7   "model_info": {  
8     "pc": {"best_model": "XGBoost", "cv_r2": 0.454},  
9     "sth": {"best_model": "XGBoost", "cv_r2": 0.533},  
10    "h2": {"best_model": "XGBoost", "cv_r2": 0.366}  
11  }  
12 }
```

B.1.2 Prediction Endpoint

Listing B.2. POST /predict request and response

```
1 POST http://178.105.34.179/api/predict  
2
```

```

3 Request:
4 {
5   "material":      "BiVO4",
6   "electrolyte":   "0.5 M Na2SO4",
7   "pH":            6.8,
8   "bias":          1.23,
9   "bandgap":       2.4,
10  "light_source":  "AM 1.5G",
11  "cocatalyst":     "FeOOH",
12  "nanostructure": "nanoparticle"
13 }
14
15 Response:
16 {
17   "status": "ok",
18   "Photocurrent": {"value": 3.215, "confidence": 0.44, "std":
19     0.312},
20   "STH":         {"value": 1.98, "confidence": 0.51, "std":
21     0.201},
22   "H2":           {"value": 42.7, "confidence": 0.39, "std":
23     5.81},
24   "performance_class": "MEDIUM",
25   "performance_score": 48.2,
26   "units": {
27     "Photocurrent": "mA/cm2",
28     "STH": "%",
29     "H2": "umol/h/cm2"
30   }
31 }

```

B.2 Running the Application

Listing B.3. Setup and launch instructions

```

1 # 1. Install dependencies
2 pip install flask flask-cors pandas numpy scikit-learn xgboost
3   joblib
4
5 # 2. Augment dataset
6 python expand_dataset.py

```

```

6
7 # 3. Train models on augmented dataset
8 cd backend
9 python train.py --data data/pec_dataset.csv --cv 5
10
11 # 4. Start API server (http://localhost:5001)
12 python app.py
13
14 # The deployed application is accessible at:
15 # http://178.105.34.179/

```

B.3 Dataset Augmentation Output

Listing B.4. `expand_dataset.py` console output

```

1 [1/4] Loading pec_dataset.csv -- 529 rows x 30 cols
2 [2/4] Filling existing rows with physics estimates...
3   Filled bandgap:          204
4   Filled pH:              114
5   Filled temperature:     420
6   Filled light intensity:  61
7   Estimated STH:          361
8   Estimated H2:           357
9 [3/4] Generating synthetic rows -- 147 rows
10 [4/4] Total rows: 676
11   STH filled:  556 (82%)
12   H2 filled:   559 (83%)
13   Bandgap:     609 (90%)
14   Data_Source: estimated=516 / synthetic=147 / original=13

```